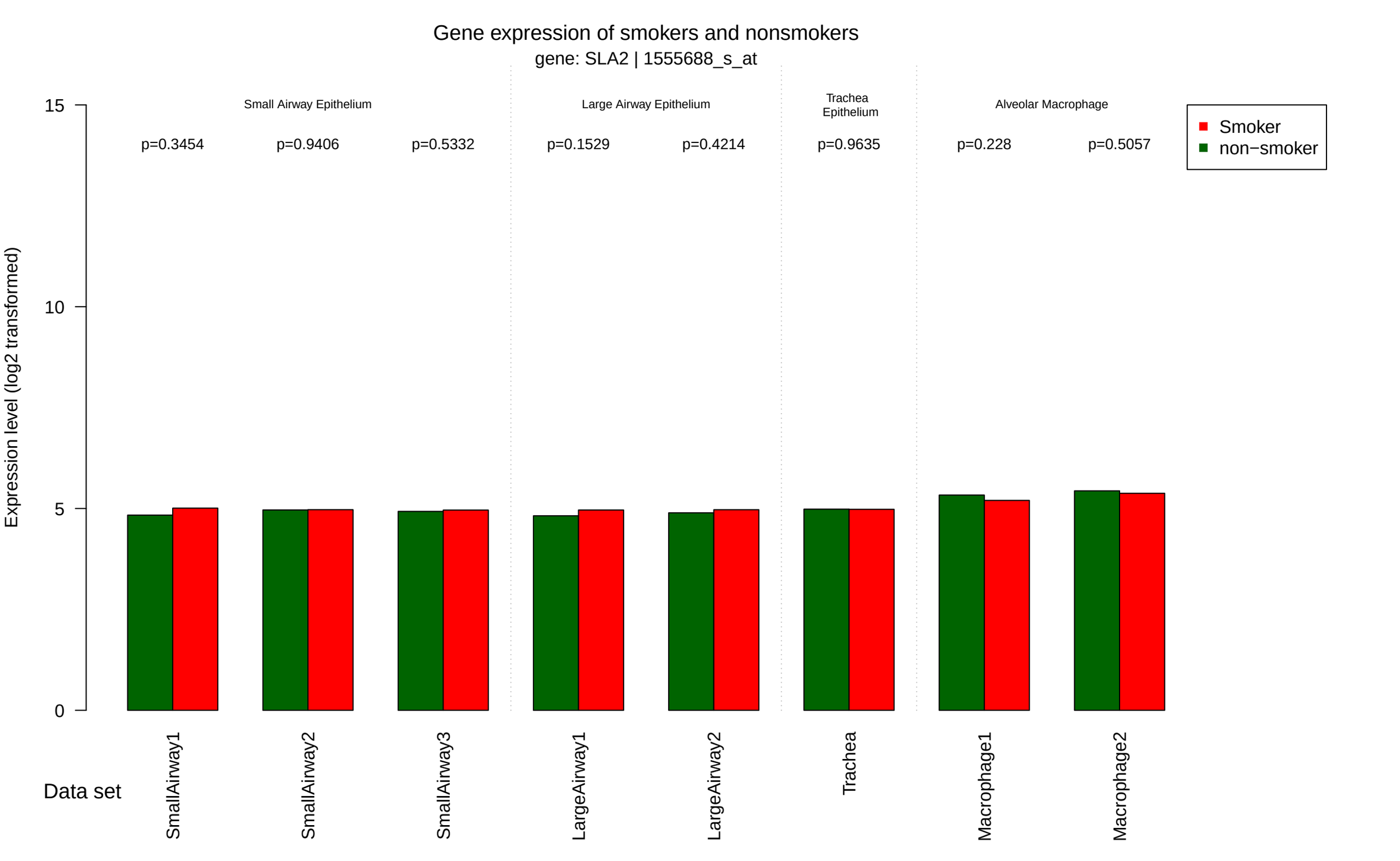
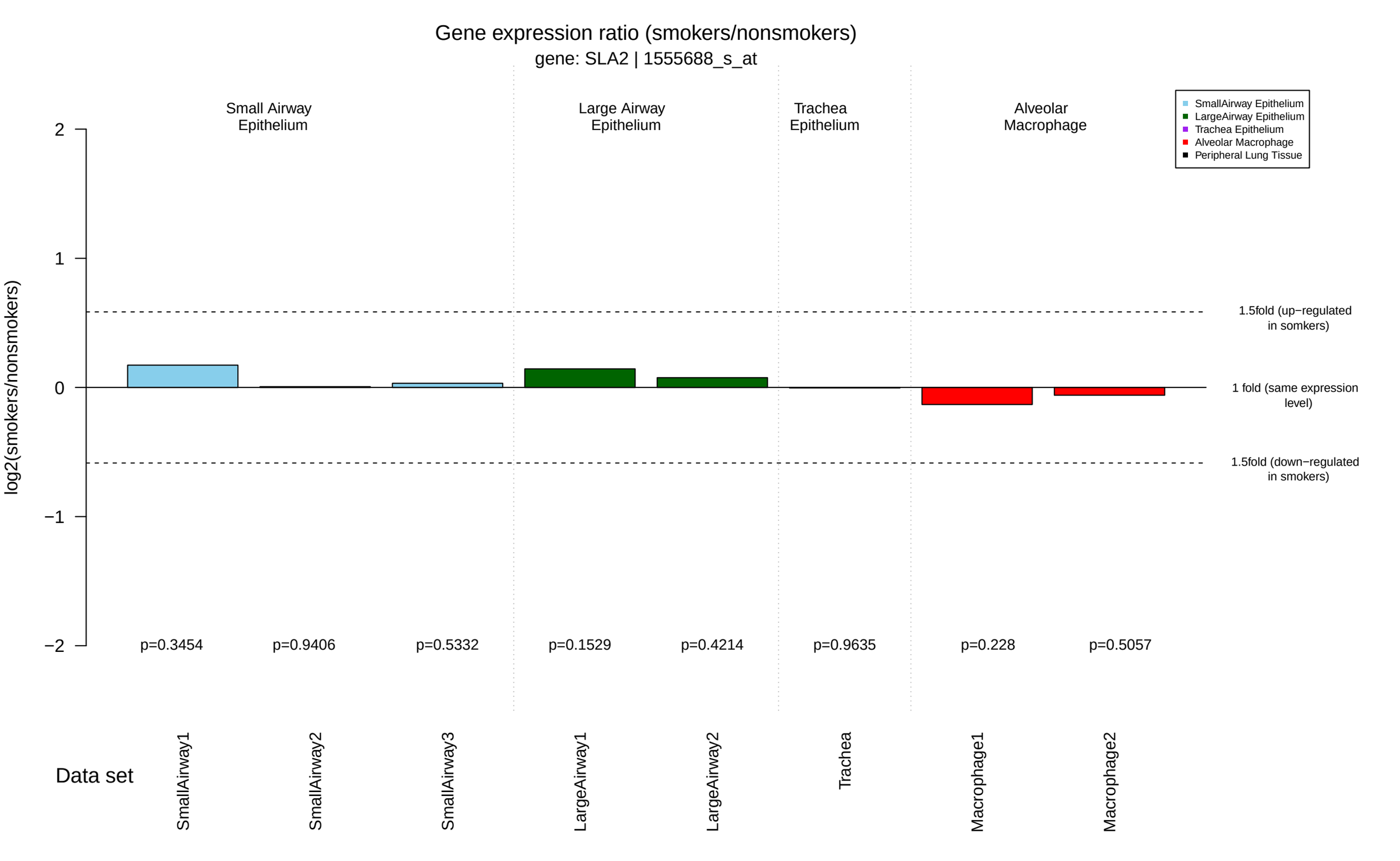
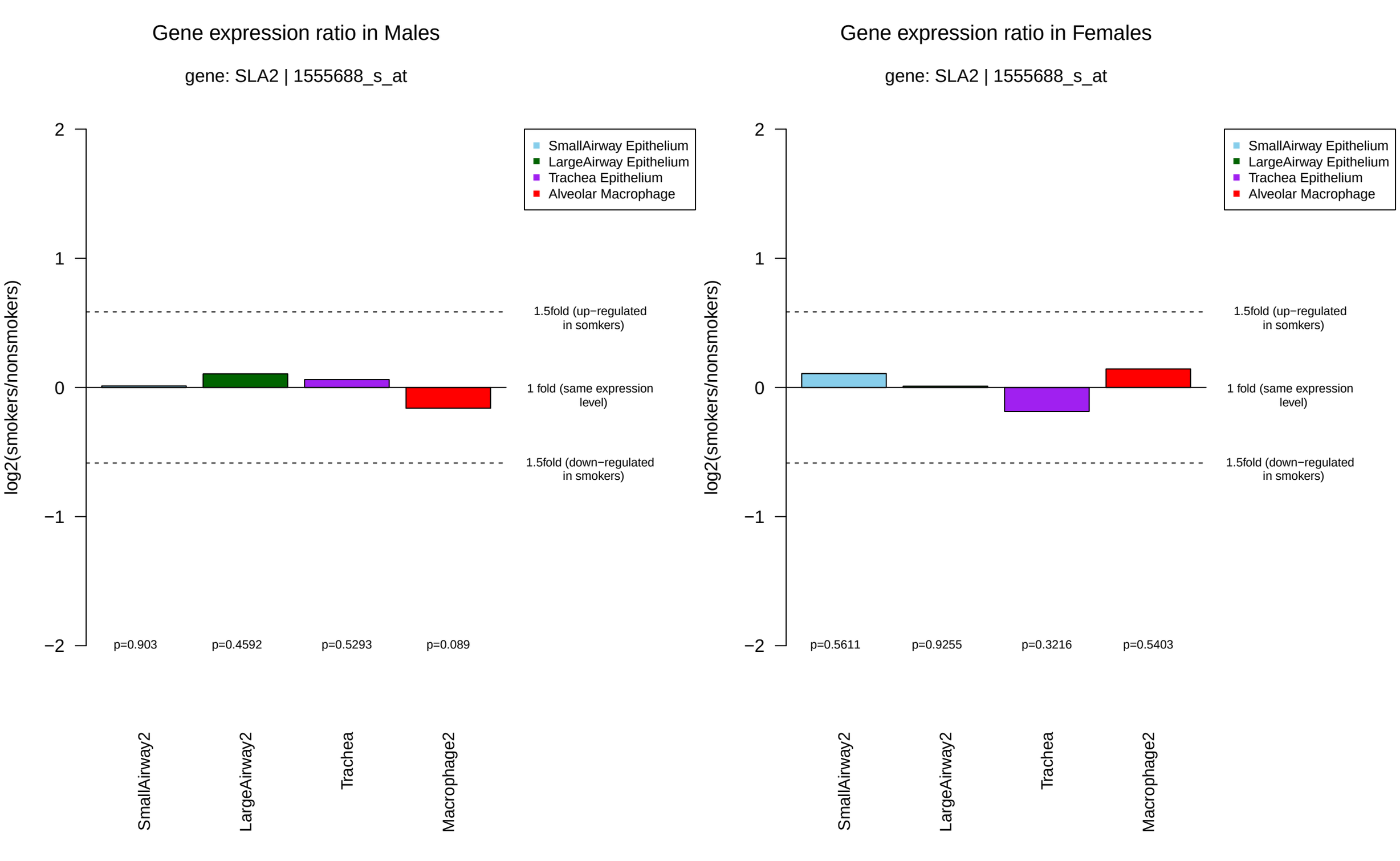
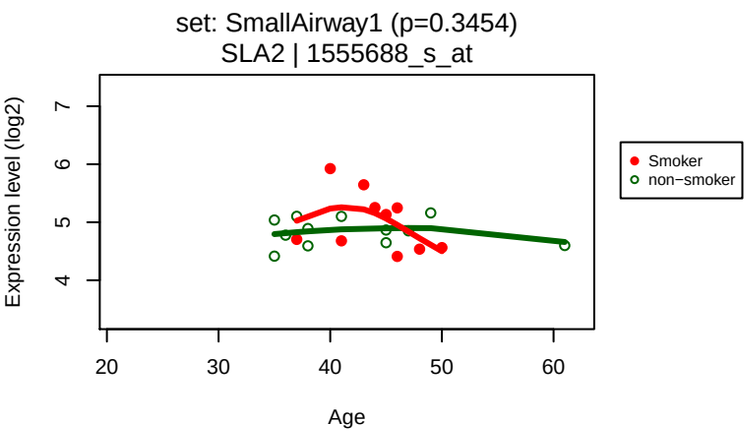
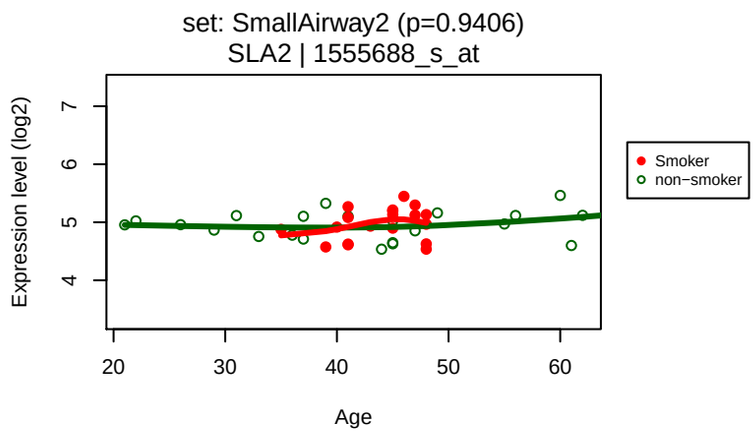
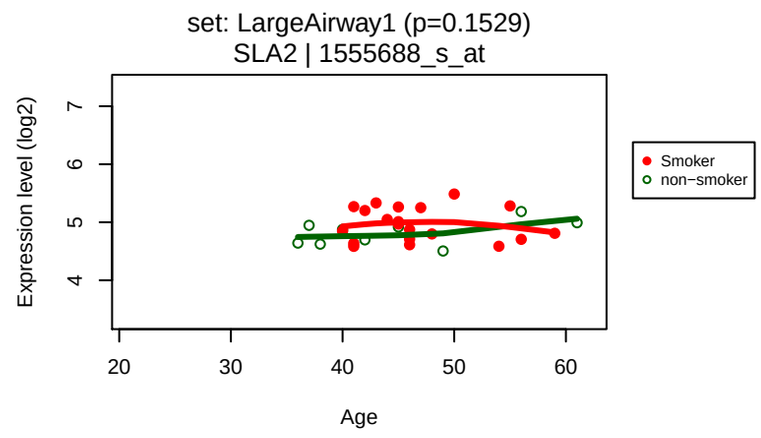

Gene expression of smokers and nonsmokers
gene: SLA2 | 1555688_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3454
p=0.9406
p=0.5332
p=0.1529
p=0.4214
p=0.9635
p=0.228
p=0.5057
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

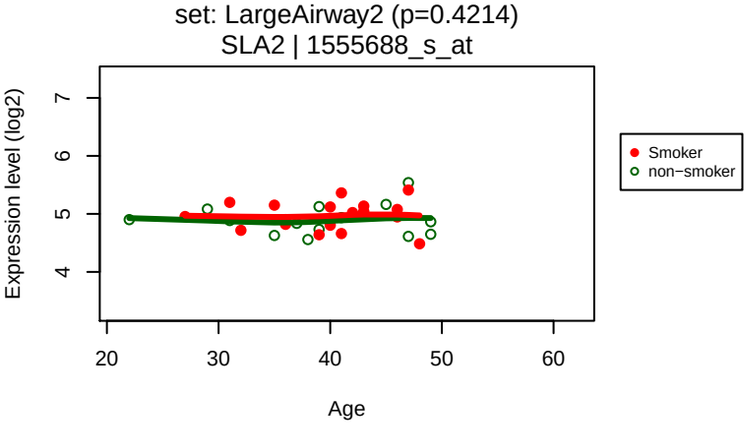

Gene expression ratio (smokers/nonsmokers)
gene: SLA2 | 1555688_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3454
p=0.9406
p=0.5332
p=0.1529
p=0.4214
p=0.9635
p=0.228
p=0.5057
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

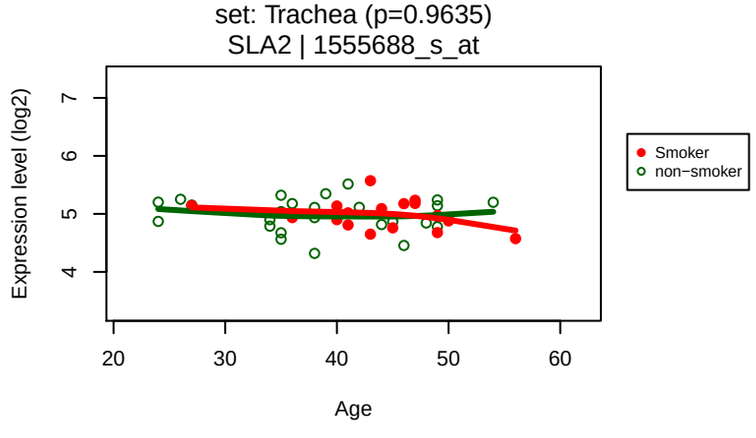

Gene expression ratio in Males
gene: SLA2 | 1555688_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.903
p=0.4592
p=0.5293
p=0.089
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SLA2 | 1555688_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5611
p=0.9255
p=0.3216
p=0.5403
SmallAirway2
LargeAirway2
Trachea
Macrophage2

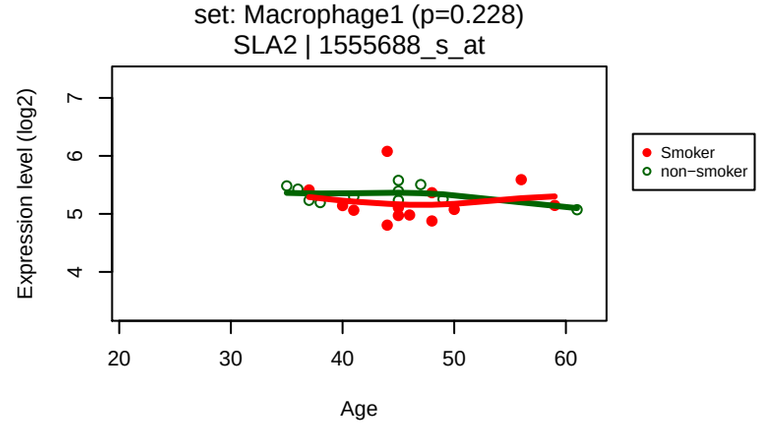

set: SmallAirway1 (p=0.3454)
SLA2 | 1555688_s_at
Expression level (log2)
Age
Smoker
non−smoker

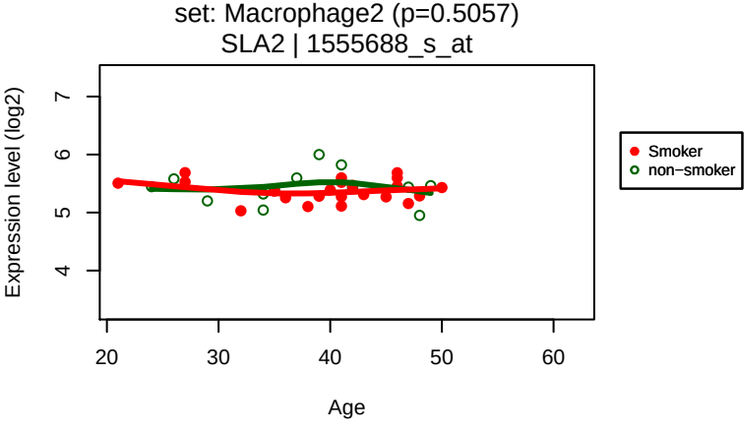

set: SmallAirway2 (p=0.9406)
SLA2 | 1555688_s_at
Expression level (log2)
Age
Smoker
non−smoker

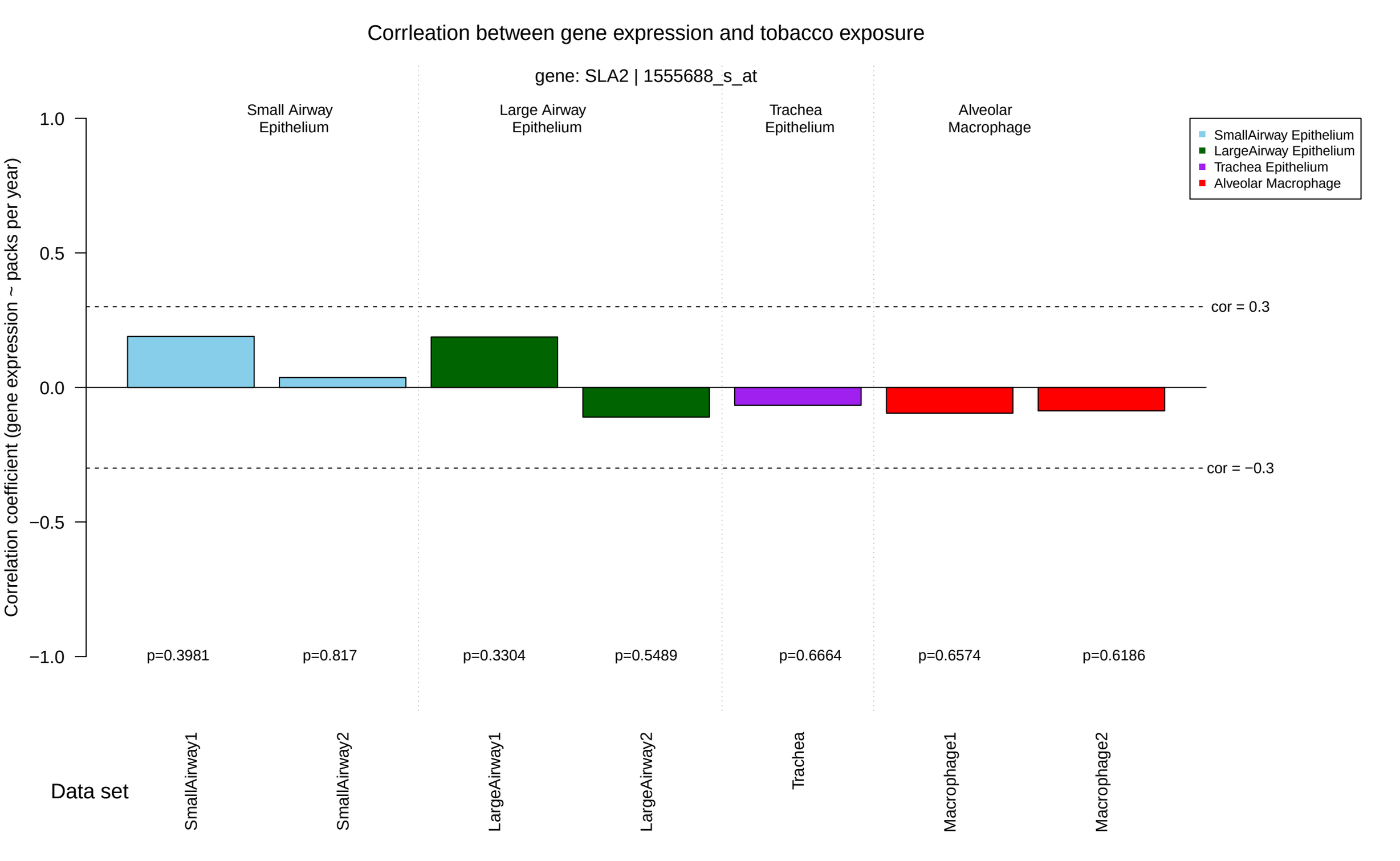

set: LargeAirway1 (p=0.1529)
SLA2 | 1555688_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4214)
SLA2 | 1555688_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9635)
SLA2 | 1555688_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.228)
SLA2 | 1555688_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5057)
SLA2 | 1555688_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SLA2 | 1555688_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3981
p=0.817
p=0.3304
p=0.5489
p=0.6664
p=0.6574
p=0.6186
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2